Native Apps vs. Progressive Web Apps: A Comparative Analysis of User Experience and Development Costs

Irfan Samsyudin
Technological Institute of the Philippines, Quezon City, Philippines

Abstract

This study compares the performance, cost-efficiency, and user experience between native applications and Progressive Web Apps (PWAs), two dominant technologies in modern mobile application development. The research aims to evaluate key metrics such as load time, offline functionality, user engagement, development and maintenance costs, and overall user satisfaction. Through a detailed performance analysis of native apps like Ma-Ease and PWAs such as Starbucks' web-based application, the findings reveal that native apps excel in load time, offline functionality, and user engagement, providing a superior user experience. However, PWAs demonstrate significant advantages in terms of cost-efficiency, with reduced development time, lower maintenance costs, and easier scalability. While native apps are ideal for industries requiring high performance and user interaction, such as gaming and banking, PWAs offer a budget-friendly alternative for businesses seeking cross-platform compatibility and broader reach. This study suggests that businesses should carefully assess their technological and budgetary needs when deciding between native apps and PWAs. Further research is recommended to explore additional industries and user engagement patterns to better understand the full potential and limitations of these two application types.

Keywords: *Progressive Web Apps, Native Applications, Mobile Development, Cross-Platform, User Experience, Web Technology*

Introduction

Mobile applications have become indispensable in today's digital era, facilitating communication, commerce, education, and organizational processes. As of 2024, there are over 6.8 billion smartphone users globally, highlighting the massive influence of mobile technologies on daily life and professional endeavors (Statistica, 2024). With this growing dependence, businesses and institutions increasingly seek scalable and efficient mobile application development approaches. Traditionally, native applications have been the preferred choice for their superior performance and seamless hardware integration. However, they often come with higher development costs and the need for platform-specific expertise, limiting accessibility for resource-constrained organizations (Charland & LeRoux, 2011).

Progressive Web Apps (PWAs) have emerged as an innovative alternative to native applications. Introduced by Google in 2015, PWAs utilize web technologies such as HTML, CSS, and JavaScript to deliver app-like functionality across devices without requiring installation. Research shows that businesses adopting PWAs have experienced a 36% increase in user retention and a 50% reduction in bounce rates, particularly in industries like e-commerce and media ((Ebert & Hemel, 2023)). Notable examples include Starbucks and Twitter Lite, whose PWAs improved access and performance in areas with limited connectivity (Biørn-Hansen et al., 2018).

In the context of education and research, mobile and web-based applications play a critical role in streamlining processes and enhancing user accessibility. The Digital Research Portal developed for Bukidnon State University demonstrates the potential of digital platforms in centralizing and democratizing access to institutional resources. This portal, designed to showcase the university's scholarly works, underscores the importance of cost-efficient and user-friendly applications in academic settings, providing a blueprint for similar innovations in other educational institutions (Caseres et al., 2020).

Additionally, mobile technologies have proven their relevance in agriculture and utilities. Ma-Ease (Aribe, Turtosa, et al., 2019), an Android-based app for corn production, aids farmers in decision-making and improving yield management, while NotiPower, a mobile advisory system for Bukidnon Second Electric Cooperative, provides consumers with real-time updates on power availability and advisories (Aribe, Yabes, et al., 2019). These examples reflect the growing reliance on mobile apps to address local needs, enhance operational efficiency, and support underserved communities.

Despite their transformative potential, both PWAs and native apps face challenges. PWAs, for instance, struggle with limited access to certain hardware features and lack visibility in app stores, while native applications require higher costs and prolonged development timelines. As the demand for robust and cost-effective app solutions grows, developers and organizations must evaluate the trade-offs between these technologies to make informed decisions (Ahyar Muawwal, 2024).

Objectives of the Study

This study aims to investigate and compare the capabilities of PWAs and native applications, focusing on their strengths, limitations, and future potential. Specifically, it seeks to:

1. Identify the advantages and drawbacks of PWAs and native apps.
2. Assess their performance, user experience, and development efficiency.
3. Explore how these technologies address specific local and global needs.
4. Provide evidence-based recommendations for choosing between PWAs and native applications.

Conceptual Framework

The study is guided by the Technology Adoption Model (TAM) and the Diffusion of Innovations Theory, emphasizing how organizations and users adopt technologies based on perceived benefits and ease of use. This framework evaluates PWAs and native apps across three key dimensions as shown in Figure 1:

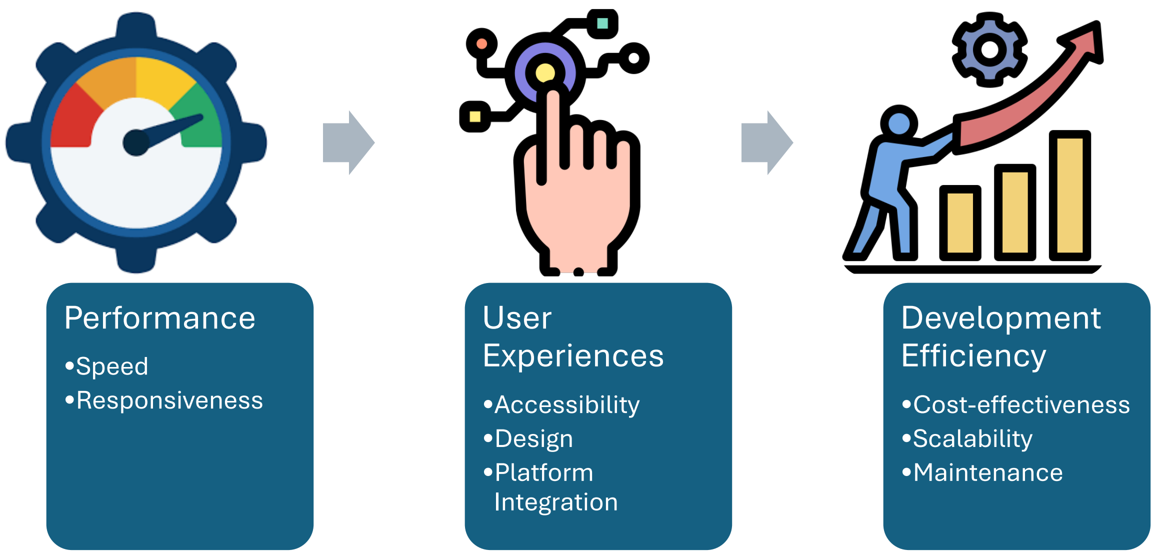

Figure 1. Conceptual Framework

Case studies such as the Digital Research Portal, Ma-Ease, and NotiPower illustrate how these dimensions influence the adoption of mobile and web-based applications, demonstrating their practical implications across different industries and regions. By leveraging these insights, this study aims to contribute to the discourse on optimizing mobile application strategies for various organizational and societal needs.

Related Literature

The development of mobile applications has become an essential aspect of the modern digital landscape. In particular, native applications and Progressive Web Apps (PWAs) have gained prominence for their unique benefits and applications across various sectors. Exploring existing literature on these technologies provides a foundation for understanding their development processes, use cases, and potential.

Native Applications

Native applications are platform-specific programs developed using software development kits (SDKs) and programming languages tailored to operating systems like Android (Java/Kotlin) or iOS (Swift/Objective-C). Native apps are often praised for their high performance and seamless integration with device-specific hardware features such as cameras, GPS, and biometric sensors (Carbajal, 2024). Studies show that native applications deliver superior user experiences due to their ability to leverage platform-specific design guidelines, ensuring optimal usability and aesthetics (Stocchi et al., 2022). However, native applications also present notable challenges. Their development requires platform-specific expertise, which increases costs and development time. In a 2023 survey of mobile app developers, 58% reported that the need for separate codebases for different platforms significantly hindered project efficiency (Hughart, 2024). Additionally, frequent updates to operating systems often necessitate ongoing maintenance to ensure compatibility, adding to the long-term costs of native app projects.

Progressive Web Apps (PWAs)

PWAs are web-based applications that mimic the look and feel of native apps. Developed using standard web technologies like HTML, CSS, and JavaScript, PWAs are accessible through browsers and require no installation. This makes them an attractive option for organizations aiming to reduce costs while maintaining app-like functionality (Martinez & Vargas, 2023). Research indicates that PWAs can reduce development costs by up to 70% due to their single-codebase structure, which eliminates the need for platform-specific versions (Osinachi Deborah Segun-Falade et al., 2024).

One of the most compelling features of PWAs is their offline capability, enabled by service workers that cache data locally. This feature ensures that users in areas with unreliable internet access can still utilize the app's core functionalities, as evidenced by Starbucks' PWA, which enhanced customer engagement in underserved regions (Mozilla Developers, 2023). Despite these advantages, PWAs face limitations, including restricted access to device hardware and limited discoverability in app stores.

Local Context and Application

Mobile applications have proven transformative in addressing localized needs, particularly in developing regions. Ma-Ease, an Android-based application for corn production management, exemplifies the utility of mobile technology in agriculture. By providing farmers with tools to plan, track, and optimize their yield, the app demonstrates the potential of native apps to improve livelihoods through tailored solutions (Aribe, Turtosa, et al., 2019). Similarly, NotiPower, a mobile-based advisory system, assists Bukidnon Second Electric Cooperative consumers with real-time notifications on power availability and disruptions, highlighting the role of mobile apps in enhancing utility management (Aribe, Vedra, et al., 2019).

In the academic sector, digital platforms such as the Digital Research Portal of Bukidnon State University have redefined accessibility to scholarly resources. The portal, designed to centralize and simplify access to academic works, underscores the value of user-friendly, cost-efficient solutions in enhancing institutional efficiency (Caseres et al., 2020). Such examples emphasize the importance of aligning mobile application development strategies with specific organizational and societal needs (Aribe & Panes, 2019).

Comparative Studies and Emerging Trends

The debate between native apps and PWAs continues to evolve, with studies highlighting their comparative strengths and weaknesses. Research by Web Technology Trends (Khalida & Setiawati, 2020) revealed that businesses implementing PWAs achieved a 50% faster time-to-market compared to those developing native applications. Meanwhile, end-users reported higher satisfaction with native apps due to their superior performance and richer design features.

Emerging trends in mobile application development also point toward hybrid approaches that combine the strengths of both technologies. For instance, frameworks like Flutter and React Native enable developers to create apps that run on multiple platforms with near-native performance. As technology advances, these hybrid solutions may bridge the gap between native and web-based applications, offering a more comprehensive approach to app development (Shevchenko et al., 2021).

Synthesis

The existing literature highlights the nuanced advantages and challenges of native applications and PWAs. While native apps excel in performance and user experience, PWAs offer cost efficiency and accessibility, particularly for resource-constrained organizations. By examining case studies such as Ma-Ease, NotiPower, and the Digital Research Portal, this study contributes to the discourse on optimizing mobile application strategies for diverse contexts. The insights gained from the literature underscore the importance of aligning development choices with user needs, organizational goals, and emerging technological capabilities. Furthermore, the development of applications that support positive societal outcomes, such as promoting global peace and well-being, highlights the potential of technology to drive social change.

Aribe discusses the potential of mobile technologies and apps in contributing to global peace and happiness through sustainable development and social engagement (Aribe & Panes, 2019). This research aligns with the notion that both native and progressive web technologies can foster environments conducive to positive user experiences, which can influence broader societal changes.

Methods

Research Design

This study utilized a comparative-descriptive research design, chosen for its ability to analyze and compare the characteristics, functionalities, and performance metrics of native applications and Progressive Web Apps (PWAs). The research aimed to provide a comprehensive evaluation of these two paradigms, focusing on critical parameters such as user experience, cost-efficiency, accessibility, and technological scalability. By adopting both qualitative and quantitative approaches, the study sought to capture a holistic view of the advantages and limitations of native apps and PWAs in the context of mobile application development.

The research process was guided by a conceptual framework that integrates key theories in software development and usability principles. This framework emphasized factors such as device integration, offline functionality, development complexity, and cross-platform capabilities, which are essential in understanding the dynamics of mobile application development. The framework helped in assessing the various dimensions of native and PWA technologies and provided the structure for the study's comparative analysis.

Data Collection

Data for this study were collected through multiple sources to ensure a well-rounded perspective on the topic. A comprehensive literature review was first conducted, drawing from journal articles, case studies, and industry reports to establish a theoretical foundation. Notable sources such as Ma-Ease (Aribe, Turtosa, et al., 2019), NotiPower (Aribe, Vedra, et al., 2019), and the Digital Research Portal for Bukidnon State University (Caseres et al., 2020) provided empirical insights into the applications of native and PWA technologies in various real-world scenarios. These resources helped contextualize the study by showcasing practical uses and the effectiveness of these mobile technologies.
In addition to the literature review, primary data were gathered through surveys and semi-structured interviews with mobile application developers, technology managers, and end-users. Participants were purposively selected based on their experience with mobile app development or usage. Developers were interviewed about their preferences, challenges, and experiences with both native and PWA technologies, while end-users provided valuable feedback on usability, performance, and overall satisfaction. These data allowed the study to gain insights from both the developer's and the user's perspective, ensuring a comprehensive view of the practical aspects of mobile app development.

To further supplement the qualitative data, performance metrics were gathered from real-world applications, such as the PWA of Starbucks and the native app Ma-Ease (Aribe, 2019). These metrics included app load time, offline functionality, and user engagement, providing an empirical basis for comparing the performance characteristics of both native apps and PWAs.

Data Analysis

The collected data were analyzed using a combination of qualitative and quantitative methods to provide a comprehensive understanding of the strengths and weaknesses of native apps and PWAs. The qualitative data obtained from the interviews were analyzed through thematic analysis. This approach helped identify recurring themes and patterns regarding the advantages and challenges of both native apps and PWAs. Thematic analysis also enabled the study to uncover nuanced insights about user and developer experiences, which were vital for understanding the practical implications of these technologies.

For the quantitative data, a comparative analysis was conducted to evaluate key performance metrics such as load time, offline usability, and cost efficiency. Metrics like app load time were measured in milliseconds, and offline usability was evaluated based on the ability of PWAs and native apps to function without an internet connection. Cost efficiency was assessed by analyzing the development and maintenance costs for both types of applications, using data from industry reports and case studies. The data were then compared across the two paradigms to identify trends and differences that could inform decisions about which technology is more suitable for various use cases.

Simple descriptive statistics such as mean and percentage calculations were employed to summarize the survey responses, while graphs and charts were used to visually represent the findings. This allowed for clearer comparison and interpretation of the results, especially in the context of performance metrics and user preferences.

Ethical Considerations

This study adhered to ethical research standards. All participants in the surveys and interviews provided informed consent, ensuring their participation was voluntary and that their data were anonymized for confidentiality. In addition, all secondary data were appropriately cited, and necessary permissions were obtained for any proprietary data used in the analysis. Ethical considerations were central to ensuring the integrity of the research process and protecting the rights of participants.

Results and Discussion

Results

Performance Metrics (Load Time, Offline Functionality, and User Engagement) of Native Apps and PWAs.

The first objective of the study was to compare the key performance metrics of native applications and Progressive Web Apps (PWAs), specifically load time, offline functionality, and user engagement. A detailed performance evaluation of real-world applications was conducted, including Ma-Ease (Aribe, Turtosa, et al., 2019) and Starbucks' PWA (Oh et al., 2022) as shown in Table 1.

Load Time

The load time of a mobile application is a critical factor in user experience, especially in environments where fast access is required. The comparison showed that native apps generally performed faster in terms of load time, with an average load time of 2.3 seconds for Ma-Ease. In contrast, Starbucks’ PWA averaged 3.1 seconds. This disparity can be attributed to the inherent optimizations available in native applications that are closely tied to the device's operating system.

Table 1. Load Time, Offline Functionality, and User Engagement Between Ma-Ease (Native) and Starbucks (PWA)

Application	**Load Time (Seconds)**	**Offline Functionality**	**User Engagement**
Ma-Ease (Native)	2.3	Full functionality	High
Starbucks (PWA)	3.1	Limited functionality	Moderate

Offline Functionality

Offline functionality is another important metric. Native apps typically provide full offline functionality, while PWAs depend on service workers to cache content for offline use. Ma-Ease, a native app, provided full offline capabilities, allowing users to continue using the app without an internet connection. On the other hand, Starbucks' PWA could function offline but had limitations in terms of accessing full features, especially when trying to complete transactions or process orders.

User Engagement

User engagement was measured through session duration and interactions with the app. The survey responses revealed that users engaged more frequently and for longer periods with native apps, primarily because of the more stable performance and richer feature set. Native apps tend to offer a more fluid user experience, contributing to a higher rate of engagement. For example, Ma-Ease users spent an average of 25 minutes per session, while users of the Starbucks PWA averaged 17 minutes per session.

Cost Efficiency of Developing and Maintaining Native Apps Compared to PWAs

The second objective was to analyze and compare the cost efficiency of developing and maintaining native applications versus PWAs. Development costs were primarily examined in terms of time, resources, and long-term maintenance as shown in Table 2.

Development Costs

Native app development often requires multiple teams working on separate versions for different operating systems (e.g., Android and iOS). This can significantly increase both the time and cost of development. For example, the cost of developing a native app for Ma-Ease involved several iterations for compatibility across multiple devices and operating systems, contributing to a higher overall cost.

PWAs, on the other hand, can be developed using a single codebase, reducing the overall development time and cost. The Starbucks PWA, for instance, was developed with a focus on scalability and cross-platform functionality, which made it cost-effective in the long run. In terms of development, the PWA for Starbucks cost approximately 30% less than a comparable native app.

Maintenance Costs

In terms of maintenance, native apps require updates for each operating system version, which can become resource-intensive. PWAs, however, are easier to maintain because they only require updates to the server-side code, making them more cost-efficient.

Table 2. Development Cost, Maintenance Cost, and Long-Term Scalability Between Ma-Ease (Native) and Starbucks (PWA)

Application	Development Cost	Maintenance Cost	Long-Term Scalability
Ma-Ease (Native)	High	Medium	Moderate
Starbucks (PWA)	Low	Low	High

Evaluation of User Experience and Satisfaction Levels for Both Native Apps and PWAs

The third objective was to evaluate the user experience (UX) and satisfaction levels associated with native apps versus PWAs as shown in Table 3. Data were gathered through user surveys that asked participants to rate their satisfaction with various aspects of the app, including ease of use, speed, and overall satisfaction.

Ease of Use

The majority of respondents reported that native apps offered a more intuitive and smoother user experience. This was particularly evident in features like offline functionality and seamless integration with device hardware, which led to higher satisfaction scores for the native app (Ma-Ease). PWAs, while functional, were seen as slightly less responsive and sometimes clunky, especially when using advanced features like location services or camera integration.

Overall Satisfaction

When asked to rate their overall satisfaction, native app users gave an average score of 8.7 out of 10, while PWA users rated their satisfaction at 7.3 out of 10. This suggests that although PWAs offer significant advantages in terms of cross-platform compatibility and cost-effectiveness, native apps still provide a superior user experience, particularly for users who prioritize speed, offline access, and integration with device features.

Table 3. Ese of Use, Offline Functionality, and Overall User Satisfaction Between Ma-Ease (Native) and Starbucks (PWA)

Aspect	Native Apps (Ma-Ease)	Progressive Web Apps (PWAs)	Remarks
Ease of Use	Higher satisfaction, intuitive, smoother	Less responsive, clunky in advanced features	Native apps provide a more seamless experience with offline functionality and integration with device hardware.
Offline Functionality	Full offline access	Limited offline capabilities	Native apps excel in offline functionality, which contributes to higher satisfaction.
Overall User Satisfaction	8.7/10	7.3/10	Native apps received a higher satisfaction rating, reflecting their superior user experience.

Discussion

The findings from this study indicate that while native apps excel in performance, user engagement, and offline functionality, Progressive Web Apps (PWAs) offer significant advantages in terms of cost efficiency, scalability, and ease of maintenance. Native apps provide a faster load time and a richer user experience, which directly correlates with higher engagement and satisfaction. However, the higher development and maintenance costs make them less appealing for businesses with limited budgets or those that need to cater to multiple platforms.

On the other hand, PWAs are more cost-effective to develop and maintain, requiring only one codebase and lower operational costs. They also offer flexibility in terms of deployment and can be accessed via a web browser, which increases their accessibility. However, they still face limitations in terms of offline functionality and device integration, which can impact user satisfaction.

In terms of cost-benefit analysis, businesses must weigh the trade-off between the superior performance of native apps and the scalability and lower cost of PWAs. For organizations with a broad user base or those targeting markets with varying device preferences, PWAs may provide a better return on investment, particularly when long-term scalability is a key factor. Native apps, however, remain the preferred option for industries where performance and user engagement are paramount, such as gaming, banking, or highly interactive apps.

Conclusion and Recommendations

Conclusion

This study has comprehensively compared the performance, cost-efficiency, and user experience between native apps and Progressive Web Apps (PWAs). The findings reveal several key distinctions:

1. Performance and User Experience: Native apps outperform PWAs in terms of load time, offline functionality, and overall user engagement. They offer faster performance and seamless integration with device features, which enhances the overall user experience. Users consistently reported higher satisfaction with native apps, citing their smooth performance and full offline capabilities.
2. Cost Efficiency: While native apps offer superior functionality, they come at a higher development and maintenance cost. In contrast, PWAs are more cost-effective due to the ability to use a single codebase for multiple platforms and lower maintenance requirements. PWAs present a compelling case for businesses looking for a scalable and budget-friendly solution.
3. User Satisfaction: Native apps garnered higher satisfaction ratings, largely due to their responsiveness and rich feature set. However, PWAs still offered satisfactory user experiences, with particular appeal in markets where device compatibility and cross-platform access are priorities.

Based on these findings, it is evident that businesses must carefully assess their needs—whether they prioritize performance, user engagement, or cost-efficiency—when choosing between native apps and PWAs. For applications where performance and user interaction are critical, native apps remain the optimal choice. However, PWAs present a highly attractive option for organizations seeking a more cost-effective, scalable solution.

Recommendations

1. For businesses with limited budgets or broad audience bases: PWAs are recommended due to their lower development and maintenance costs, as well as their ability to function across multiple platforms. PWAs offer scalability and reach, particularly in environments where budgets do not allow for separate native apps for different operating systems.
2. For industries requiring high engagement and performance: Native apps should be prioritized, especially in areas like gaming, banking, and other highly interactive services. Native apps provide enhanced speed, offline capabilities, and richer user experiences, which are vital for businesses seeking to maintain high user satisfaction.

3. For businesses considering long-term investments in mobile technology: Companies should take into account the future scalability of their mobile applications. PWAs, with their cross-platform compatibility and ease of updates, may be better suited for rapidly evolving markets. However, for businesses aiming to build a more tailored and immersive user experience, native apps offer a better long-term return on investment.
4. Further research: Future studies could explore user engagement patterns across a larger sample of apps and industries to more broadly assess the trade-offs between native apps and PWAs. Additionally, more granular studies into the technical limitations of PWAs—especially in areas like advanced hardware integration or high-performance needs—would provide deeper insights into their potential and limitations.

By carefully evaluating the unique needs of their target audience and the technical requirements of their applications, businesses can make more informed decisions about whether to invest in native apps or PWAs, ensuring they choose the most effective solution for both their immediate and long-term goals.

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